

The centre of the even hybrid family quantum GL_1

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Abstract

A family quantum group carries, alongside the deformation parameter, independent toral parameters $t_1, \dots, t_n$. Its *even hybrid integral form* is the part of it that preserves every integral Verma lattice. This paper determines the centre of that form in rank one. With the root system empty the Harish–Chandra projection is the identity, and the content is a statement about Laurent polynomials on a geometric grid: the centre is exactly the *even Newton lattice* of those taking values in A_t at every weight, and it is spanned over A_t by the shifted divided classes $\check{Y}^u \nu_r(\check{Y})$. This is a bilateral form of the quantum Pólya theorem of Harman and Hopkins [5]. The rank-one case is the first instance of a corresponding statement for GL_n , which is the subject of work in progress. Sections 2 and 3 state and prove the rank-one theorem; Section 4 is a short history.

1 Introduction

Two things distinguish the family quantum group from the usual one. The toral parameters $t_1, \dots, t_n$ are free, so the algebra is a family over a torus rather than a single object; and one takes not the whole algebra but the *hybrid integral form* — the elements preserving every integral Verma lattice — and inside that, the torally even part. The centre of that form is the object of study.

The mechanism is worth saying before any notation. A central element is determined by what it does to a highest-weight vector, weight by weight; that assignment is a function on a grid of weights; and the condition of being *integral* — of landing in A_t at every weight rather than merely in the fraction field — cuts out a lattice of such functions. *The theorem below is that in rank one the centre is exactly that lattice.* Nothing is lost and nothing extra appears.

The lattice in question has a long classical ancestry. That the integer-valued polynomials on $\mathbb{Z}$ are free on the binomial coefficients is Pólya’s basis theorem [5, Prop. 1.1]; the standard reference for the subject is Cahen–Chabert [1]. On a *geometric* progression — the grid that appears here — the corresponding basis is Gramain’s [3, Prop. 2.2], for q a natural number; the q -analogue on the additive grid, with q an indeterminate, is Harman–Hopkins [5]. Section 4 places the present setting among them.

There is a corresponding statement for GL_n , of which the theorem below is the first instance — the case where the root system is empty. It is the subject of work in progress.

Sections 2 and 3 state and then prove the theorem in full. With $\Phi = \emptyset$ and $W = \{1\}$ the algebra is a torus, the Harish–Chandra projection is the identity, and the three clauses reduce to one substantial assertion: the even Newton lattice is spanned over A_t by the shifted divided classes $\check{Y}^u \nu_r(\check{Y})$.

2 The case $n = 1$

Here $\Phi = \emptyset$, $W = \{1\}$, and the cocharacter lattice is $\Lambda = \mathbb{Z}$. Much of the general apparatus is vacuous in this case: there are no root vectors, so the PBW basis is the toral monomials alone; the Harish–Chandra projection is the identity; and Weyl invariance imposes nothing. What survives is the integrality condition, and it is not vacuous at all.

Let $G = GL_1$: the root system is $\Phi = \emptyset$, the Weyl group is $W = \{1\}$, the cocharacter lattice is $\Lambda = \mathbb{Z}$, and $X_\Lambda = \mathbb{Z}$. Let

$$A_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}], \quad F_t = \mathbb{Q}(v)[t^{\pm 1}], \quad q = v^2,$$

with v, t independent indeterminates.

Definition 1 (Hybrid family quantum GL_1). *U is the F_t -algebra with invertible pairwise commuting generators $K_{n,+}, K_{n,-}$ ($n \in \mathbb{Z}$) subject to*

$$K_{0,\pm} = 1, \quad K_{m+n,\pm} = K_{m,\pm} K_{n,\pm}, \quad K_{n,+} K_{n,-} = t^n.$$

There are no generators e_i, F_i , since $\Phi = \emptyset$. Writing $T^\pm := K_{1,\pm}$, the monomials $(T^+)^w$ ($w \in \mathbb{Z}$) form a F_t -basis of U .

Definition 2 (Even Newton lattice). *A toral element is an element of $F_t[(T^+)^{\pm 1}]$; it is even when all its toral exponents are even — equivalently, it is a Laurent polynomial in $\check{Y} := (T^+)^2$. For $\chi \in X_\Lambda$ let ev_χ be the A_t -linear map on even toral elements determined by $\text{ev}_\chi(\check{Y}) = q^\chi$. The even Newton lattice is the A_t -submodule*

$$\mathcal{N}^{\text{ev}} = \{ f \in F_t[\check{Y}^{\pm 1}] : \text{ev}_\chi(f) \in A_t \text{ for every } \chi \in \mathbb{Z} \}.$$

Definition 3 (Even hybrid family integral form). *Since $\Phi^+ = \emptyset$, the even hybrid family integral form is the A_t -submodule*

$$U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subset U,$$

and its central part is $Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{F_t} \text{Frac}(A_t))$.

Definition 4 (Harish–Chandra projection; shifted Weyl action). *Every element of U is toral, so the Harish–Chandra projection HC is the identity map on U . With*

$$Y_n = t^{-n} K_{n,+}^2 \quad (n \in \mathbb{Z}),$$

every even toral element is a Laurent polynomial in the Y_n , and the shifted Weyl action of W is the F_t -algebra action $w(Y_n) = Y_{wn}$; set $(\mathcal{N}^{\text{ev}})^W = \{f \in \mathcal{N}^{\text{ev}} : wf = f \text{ for all } w \in W\}$.

Definition 5 (Newton divided classes). *For $r \geq 0$ set*

$$\nu_r(X) = \prod_{s=0}^{r-1} \frac{X - q^s}{q^r - q^s} \in \mathbb{Q}(v)[X], \quad \nu_0(X) = 1.$$

Statement. *For $G = GL_1$, with $\Lambda = \mathbb{Z}$:*

(i) *$U^{\text{hyb, ev}}$ is an A_t -subalgebra of U ;*

(ii) *the Harish–Chandra projection restricts to an isomorphism of A_t -algebras*

$$\text{HC} : Z(U^{\text{hyb, ev}}) \xrightarrow{\sim} (\mathcal{N}^{\text{ev}})^W;$$

(iii) *$\mathcal{N}^{\text{ev}} = \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} A_t \check{Y}^u \nu_r(\check{Y})$.*

3 Proof of the case $n = 1$

Throughout, v and t are independent indeterminates and

$$A_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}], \quad F_t = \mathbb{Q}(v)[t^{\pm 1}], \quad q = v^2.$$

Thus $A_t \subseteq F_t$ is a subring, and $\mathbb{Z}[q^{\pm 1}] = \mathbb{Z}[v^{\pm 2}] \subseteq \mathbb{Z}[v^{\pm 1}] \subseteq A_t$.

The algebra U . U is the F_t -algebra with invertible pairwise-commuting generators $K_{n,+}, K_{n,-}$ ($n \in \mathbb{Z}$) subject to $K_{0,\pm} = 1$, $K_{m+n,\pm} = K_{m,\pm} K_{n,\pm}$ and $K_{n,+} K_{n,-} = t^n$. Writing $T^{\pm} = K_{1,\pm}$, the monomials $(T^+)^w$ ($w \in \mathbb{Z}$) form a F_t -basis; U is commutative. A toral element is an element of $F_t[(T^+)^{\pm 1}]$; it is *even* when it is a Laurent polynomial in

$$\check{Y} := (T^+)^2.$$

Even toral elements thus form the Laurent-polynomial algebra $F_t[\check{Y}^{\pm 1}]$.

Evaluations. For $\chi \in \mathbb{Z}$, let $\text{ev}_{\chi}: F_t[\check{Y}^{\pm 1}] \rightarrow F_t$ be the F_t -algebra homomorphism substituting the unit $q^{\chi} \in A_t$ for $\check{Y}$, i.e.

$$\text{ev}_{\chi}\left(\sum_u c_u \check{Y}^u\right) = \sum_u c_u q^{u\chi} \quad (c_u \in F_t).$$

It is the unique A_t -linear ring homomorphism with $\text{ev}_{\chi}(\check{Y}) = q^{\chi}$, and it fixes F_t (hence A_t) pointwise. The *even Newton lattice* is

$$\mathcal{N}^{\text{ev}} = \{f \in F_t[\check{Y}^{\pm 1}] : \text{ev}_{\chi}(f) \in A_t \text{ for every } \chi \in \mathbb{Z}\}.$$

Because $\Phi^+ = \emptyset$ at GL_1 , the *even hybrid family integral form* is $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subseteq U$, with central part $Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{F_t} \text{Frac}(A_t))$. Every element of U is toral, so the Harish–Chandra projection HC is the identity map on U ; the Weyl group is $W = \{1\}$, acting trivially, so $(\mathcal{N}^{\text{ev}})^W = \mathcal{N}^{\text{ev}}$.

Newton divided classes. For $r \geq 0$,

$$\nu_r(X) = \prod_{s=0}^{r-1} \frac{X - q^s}{q^r - q^s} \in \mathbb{Q}(v)[X], \quad \nu_0(X) = 1.$$

Each ν_r has degree exactly r (its leading coefficient $\prod_{s=0}^{r-1} (q^r - q^s)^{-1}$ is a nonzero element of $\mathbb{Q}(v)$), so $\{\nu_0, \dots, \nu_d\}$ is a F_t -basis of the module $F_t[Y]_{\leq d}$ of even toral polynomials of degree $\leq d$.

We write $\mathbb{Z}[q^{\pm 1}]$ -*integral* for “lies in $\mathbb{Z}[q^{\pm 1}]$ ”. For $m \in \mathbb{Z}$ and $r \geq 0$ put

$$B_r(m) := \nu_r(q^m) = \prod_{s=0}^{r-1} \frac{q^m - q^s}{q^r - q^s}, \quad B_0(m) = 1.$$

3.1 Clause (i): $U^{\text{hyb, ev}}$ is an A_t -subalgebra

Proposition 1. $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}}$ is an A_t -subalgebra of U . (This is the bilateral, t -spectator analogue of the fact that quantum integer-valued polynomials form a ring [5, §1].)

Proof. For each $\chi \in \mathbb{Z}$ the map $\text{ev}_{\chi}: F_t[\check{Y}^{\pm 1}] \rightarrow F_t$ is a ring homomorphism, and $A_t \subseteq F_t$ is a subring; hence the preimage $\text{ev}_{\chi}^{-1}(A_t)$ is a subring of $F_t[\check{Y}^{\pm 1}]$. It contains A_t , because ev_{χ} fixes A_t pointwise, so $\text{ev}_{\chi}^{-1}(A_t)$ is an A_t -subalgebra. By definition

$$\mathcal{N}^{\text{ev}} = \bigcap_{\chi \in \mathbb{Z}} \text{ev}_{\chi}^{-1}(A_t)$$

is an intersection of A_t -subalgebras of $F_t[\check{Y}^{\pm 1}]$, hence itself an A_t -subalgebra (it contains 1 since $\text{ev}_{\chi}(1) = 1 \in A_t$, and is closed under sums, products and A_t -scalars). As $F_t[\check{Y}^{\pm 1}] \subseteq U$, this is an A_t -subalgebra of U . $\square$

3.2 Clause (ii): the Harish–Chandra isomorphism is the identity

Proposition 2. *The Harish–Chandra projection restricts to an isomorphism of A_t -algebras $\text{HC}: Z(U^{\text{hyb, ev}}) \xrightarrow{\sim} (\mathcal{N}^{\text{ev}})^W$, and this isomorphism is the identity map of $\mathcal{N}^{\text{ev}}$. (The rank-one, $\Phi^+ = \emptyset$ degeneration of the even hybrid Harish–Chandra isomorphism established at GL_2 in [9]; the classical ancestor is the Harish–Chandra isomorphism itself, whose quantum-group form the GL_2 record cites.)*

Proof. The algebra U is commutative, being spanned by monomials in the pairwise commuting invertible generators $K_{n, \pm}$. Base change is compatible with commutativity, so $U \otimes_{F_t} \text{Frac}(A_t)$ is a commutative ring and equals its own centre. Therefore

$$Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{F_t} \text{Frac}(A_t)) = U^{\text{hyb, ev}} \cap (U \otimes_{F_t} \text{Frac}(A_t)) = U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}},$$

the middle equality because $U \hookrightarrow U \otimes_{F_t} \text{Frac}(A_t)$ is injective — $F_t \hookrightarrow \text{Frac}(A_t) = \mathbb{Q}(v, t)$ and U is free over F_t — so $U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subseteq U$ is literally a subset of the base-changed algebra. Every element of U is toral, so $\text{HC} = \text{id}_U$, and its restriction to $Z(U^{\text{hyb, ev}}) = \mathcal{N}^{\text{ev}}$ is $\text{id}_{\mathcal{N}^{\text{ev}}}$. Finally $W = \{1\}$ acts trivially, so $(\mathcal{N}^{\text{ev}})^W = \mathcal{N}^{\text{ev}}$. Hence

$$\text{HC}: Z(U^{\text{hyb, ev}}) = \mathcal{N}^{\text{ev}} \xrightarrow{\text{id}} \mathcal{N}^{\text{ev}} = (\mathcal{N}^{\text{ev}})^W$$

is the identity map of the A_t -algebra $\mathcal{N}^{\text{ev}}$ (an A_t -algebra by clause (i)), which is manifestly an isomorphism of A_t -algebras. $\square$

There is no content here beyond commutativity: at GL_1 the rational centre is everything, HC is the identity, and Weyl invariance is vacuous.

3.3 Clause (iii): the Newton presentation

Write

$$S := \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} A_t \check{Y}^u \nu_r(\check{Y}) \subseteq F_t[\check{Y}^{\pm 1}]$$

for the A_t -submodule generated by the shifted divided classes $\check{Y}^u \nu_r(\check{Y})$.

Theorem 1 (Even Newton presentation for GL_1). $\mathcal{N}^{\text{ev}} = S = \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} A_t \check{Y}^u \nu_r(\check{Y})$. (The bilateral quantum Pólya theorem of Harman–Hopkins [5, §§1,4], transported to the multiplicative grid; the precise dictionary is the Attribution remark below.)

The proof occupies the rest of this section: the grid lemma, the forward inclusion $S \subseteq \mathcal{N}^{\text{ev}}$, the windowed reverse inclusion for polynomials, and the Laurent-clearing step.

3.3.1 The grid values are $\mathbb{Z}[q^{\pm 1}]$ -integral

Lemma 1. *For all $m \in \mathbb{Z}$ and $r \geq 0$ one has $B_r(m) = \nu_r(q^m) \in \mathbb{Z}[q^{\pm 1}]$ (the Gaussian-binomial value pattern of [5, §1], extended to negative nodes as their §4 bilateral localization requires). Explicitly:*

- (a) if $0 \leq m < r$ then $B_r(m) = 0$;
- (b) if $m \geq r \geq 0$ then $B_r(m) = \begin{bmatrix} m \\ r \end{bmatrix}_q \in \mathbb{Z}[q]$;
- (c) if $m = -n$ with $n \geq 1$ then

$$B_r(-n) = (-1)^r q^{-nr - \binom{r}{2}} \begin{bmatrix} n + r - 1 \\ r \end{bmatrix}_q \in \mathbb{Z}[q^{\pm 1}].$$

Proof. (a) If $0 \leq m < r$, the factor of the product at $s = m$ has numerator $q^m - q^m = 0$, so $B_r(m) = 0$.

(b) Assume $m \geq r \geq 0$. Pulling q^s out of each numerator and denominator factor,

$$B_r(m) = \prod_{s=0}^{r-1} \frac{q^s(q^{m-s} - 1)}{q^s(q^{r-s} - 1)} = \prod_{s=0}^{r-1} \frac{q^{m-s} - 1}{q^{r-s} - 1}.$$

Re-indexing by $i = r - s$ (so i runs from 1 to r),

$$B_r(m) = \prod_{i=1}^r \frac{q^{(m-r)+i} - 1}{q^i - 1} = \left[\begin{matrix} m \\ r \end{matrix} \right]_q,$$

the Gaussian binomial coefficient. It lies in $\mathbb{Z}[q]$: this is standard from the q -Pascal recurrence $\left[\begin{matrix} m \\ r \end{matrix} \right]_q = \left[\begin{matrix} m-1 \\ r-1 \end{matrix} \right]_q + q^r \left[\begin{matrix} m-1 \\ r \end{matrix} \right]_q$ with $\left[\begin{matrix} m \\ 0 \end{matrix} \right]_q = 1$, by induction on m .

(c) Let $m = -n$, $n \geq 1$. In each factor write the numerator as $q^{-n} - q^s = -q^{-n}(q^{n+s} - 1)$ and the denominator as $q^r - q^s = q^s(q^{r-s} - 1)$. Then

$$B_r(-n) = \prod_{s=0}^{r-1} \frac{-q^{-n}(q^{n+s} - 1)}{q^s(q^{r-s} - 1)} = (-1)^r q^{-nr} q^{-\sum_{s=0}^{r-1} s} \prod_{s=0}^{r-1} \frac{q^{n+s} - 1}{q^{r-s} - 1}.$$

Here $\sum_{s=0}^{r-1} s = \binom{r}{2}$. For the remaining product, the denominators give $\prod_{s=0}^{r-1} (q^{r-s} - 1) = \prod_{i=1}^r (q^i - 1)$ and the numerators give $\prod_{s=0}^{r-1} (q^{n+s} - 1) = \prod_{j=n}^{n+r-1} (q^j - 1)$, so

$$\prod_{s=0}^{r-1} \frac{q^{n+s} - 1}{q^{r-s} - 1} = \frac{\prod_{j=n}^{n+r-1} (q^j - 1)}{\prod_{i=1}^r (q^i - 1)} = \frac{[n+r-1]_q! / [n-1]_q!}{[r]_q!} = \left[\begin{matrix} n+r-1 \\ r \end{matrix} \right]_q,$$

where $[k]_q = (q^k - 1)/(q - 1)$ and the common powers of $(q - 1)$ cancel. The Gaussian binomial $\left[\begin{matrix} n+r-1 \\ r \end{matrix} \right]_q \in \mathbb{Z}[q]$ by part (b). Collecting the prefactor gives

$$B_r(-n) = (-1)^r q^{-nr - \binom{r}{2}} \left[\begin{matrix} n+r-1 \\ r \end{matrix} \right]_q \in \mathbb{Z}[q^{\pm 1}]. \quad \square$$

3.3.2 Forward inclusion $S \subseteq \mathcal{N}^{\text{ev}}$

Lemma 2. *For every $u \in \mathbb{Z}$ and $r \geq 0$, $\check{Y}^u \nu_r(\check{Y}) \in \mathcal{N}^{\text{ev}}$. Hence $S \subseteq \mathcal{N}^{\text{ev}}$. (The transported forward half of [5, §1]: q -binomials are quantum integer-valued.)*

Proof. Fix u, r and $\chi \in \mathbb{Z}$. Since ev_χ is a ring homomorphism with $\text{ev}_\chi(\check{Y}) = q^\chi$,

$$\text{ev}_\chi(\check{Y}^u \nu_r(\check{Y})) = q^{u\chi} \nu_r(q^\chi) = q^{u\chi} B_r(\chi).$$

By Lemma 1, $B_r(\chi) \in \mathbb{Z}[q^{\pm 1}] \subseteq A_t$, and $q^{u\chi}$ is a unit of A_t ; so the product lies in A_t . As χ was arbitrary, $\check{Y}^u \nu_r(\check{Y}) \in \mathcal{N}^{\text{ev}}$. The lattice $\mathcal{N}^{\text{ev}}$ is an A_t -module (indeed an A_t -algebra, clause (i)), so it contains every A_t -combination of these generators, i.e. $S \subseteq \mathcal{N}^{\text{ev}}$. $\square$

3.3.3 Reverse inclusion for polynomials: windowed Newton interpolation

Lemma 3. *Let $g \in \mathcal{N}^{\text{ev}}$ be an even toral polynomial, $g \in F_t[\check{Y}]$, of degree $\leq d$. Then $g = \sum_{r=0}^d c_r \nu_r(\check{Y})$ with all $c_r \in A_t$. In particular $g \in \sum_{r \geq 0} A_t \nu_r(\check{Y}) \subseteq S$. (Pólya's windowed-interpolation argument as q -deformed in the proof of [5, §1], run on the multiplicative grid with t a spectator.)*

Proof. Because $\{\nu_0, \dots, \nu_d\}$ is a F_t -basis of $F_t[\check{Y}]_{\leq d}$, there are unique $c_0, \dots, c_d \in F_t$ with

$$g = \sum_{r=0}^d c_r \nu_r(\check{Y}).$$

Evaluate on the window $\chi = m$ for $m = 0, 1, \dots, d$. As ev_m is F_t -linear and $\text{ev}_m(\nu_r(\check{Y})) = B_r(m)$,

$$\text{ev}_m(g) = \sum_{r=0}^d B_r(m) c_r = \sum_{r=0}^d V[m, r] c_r, \quad V[m, r] := B_r(m) \quad (0 \leq m, r \leq d).$$

By Lemma 1, every entry $V[m, r] \in \mathbb{Z}[q^{\pm 1}]$; moreover V is *lower unitriangular*: $V[m, r] = B_r(m) = 0$ for $m < r$ (part (a)), and $V[m, m] = B_m(m) = \prod_{s=0}^{m-1} \frac{q^m - q^s}{q^m - q^s} = 1$. Hence $\det V = 1$, and the inverse V^{-1} is again lower unitriangular with entries in $\mathbb{Z}[q^{\pm 1}]$ (back-substitution never leaves $\mathbb{Z}[q^{\pm 1}]$, and by uniqueness this inverse coincides with the inverse taken over the larger ring F_t).

Writing $\mathbf{c} = (c_0, \dots, c_d)^\top$ and $\mathbf{v} = (\text{ev}_0(g), \dots, \text{ev}_d(g))^\top$, we have $\mathbf{v} = V\mathbf{c}$ in F_t^{d+1} , hence

$$\mathbf{c} = V^{-1}\mathbf{v}.$$

Since $g \in \mathcal{N}^{\text{ev}}$, each $\text{ev}_m(g) \in A_t$, so $\mathbf{v} \in A_t^{d+1}$; and V^{-1} has entries in $\mathbb{Z}[q^{\pm 1}] \subseteq A_t$. Therefore $\mathbf{c} = V^{-1}\mathbf{v} \in A_t^{d+1}$, i.e. every $c_r \in A_t$.

The t -spectator argument. The map ev_m substitutes only $\check{Y} \mapsto q^m = v^{2m}$; it does not touch v or t . Thus t never enters the matrix V , whose entries lie in $\mathbb{Z}[q^{\pm 1}]$. Concretely, expand each coordinate along the free $\mathbb{Z}[v^{\pm 1}]$ -basis of t -powers: $c_r = \sum_k c_{r,k} t^k$ with $c_{r,k} \in \mathbb{Q}(v)$, so that $\text{ev}_m(g) = \sum_k (\sum_r B_r(m) c_{r,k}) t^k$. Membership $\text{ev}_m(g) \in A_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}]$ means that for each fixed k the t^k -coefficient $\sum_r B_r(m) c_{r,k}$ lies in $\mathbb{Z}[v^{\pm 1}]$ for all $m \in \{0, \dots, d\}$. For each fixed k this is precisely the unitriangular system above, over $\mathbb{Z}[q^{\pm 1}]$ with right-hand sides in $\mathbb{Z}[v^{\pm 1}]$; it gives $c_{r,k} \in \mathbb{Z}[v^{\pm 1}]$. Hence $c_r = \sum_k c_{r,k} t^k \in \mathbb{Z}[v^{\pm 1}, t^{\pm 1}] = A_t$, as claimed. This is the same conclusion reached above; we spell out the coefficient-wise reading to make the spectator role of t explicit.

Base change $\mathbb{Z}[q^{\pm 1}] \rightarrow \mathbb{Z}[v^{\pm 1}] \rightarrow A_t$. Two rings appear: the matrix V and its inverse live over $\mathbb{Z}[q^{\pm 1}] = \mathbb{Z}[v^{\pm 2}]$, whereas the coordinates c_r and the values $\text{ev}_m(g)$ live over $F_t = \mathbb{Q}(v)[t^{\pm 1}]$ — honest v (odd powers) may occur, so the c_r need not lie in $\mathbb{Q}(q)$. This causes no loss: the extension $\mathbb{Z}[q^{\pm 1}] \hookrightarrow \mathbb{Z}[v^{\pm 1}]$ is free with basis $\{1, v\}$, hence flat, and multiplication by a $\mathbb{Z}[q^{\pm 1}]$ -entry preserves each of the two grades (even/odd powers of v). The reverse computation $\mathbf{c} = V^{-1}\mathbf{v}$ therefore runs identically in each grade, so integrality of the values over $\mathbb{Z}[v^{\pm 1}, t^{\pm 1}]$ forces integrality of the coordinates over the same ring. Equivalently, and most directly: V^{-1} has entries in $\mathbb{Z}[q^{\pm 1}] \subseteq A_t$ and $\mathbf{v} \in A_t^{d+1}$, so $\mathbf{c} = V^{-1}\mathbf{v} \in A_t^{d+1}$. $\square$

3.3.4 Laurent clearing and conclusion

Proof of Theorem 1. The inclusion $S \subseteq \mathcal{N}^{\text{ev}}$ is Lemma 2. For $\mathcal{N}^{\text{ev}} \subseteq S$, let $f \in \mathcal{N}^{\text{ev}} \subseteq F_t[\check{Y}^{\pm 1}]$ be arbitrary. Choose $N \geq 0$ so large that $g := \check{Y}^N f \in F_t[\check{Y}]$ is an honest polynomial (take N at least the largest magnitude of a negative $\check{Y}$ -exponent occurring in f). For every $\chi \in \mathbb{Z}$, using that ev_χ is a ring homomorphism,

$$\text{ev}_\chi(g) = \text{ev}_\chi(\check{Y}^N) \text{ev}_\chi(f) = q^{N\chi} \text{ev}_\chi(f).$$

Here $q^{N^\times}$ is a unit of A_t and $\text{ev}_\chi(f) \in A_t$, so $\text{ev}_\chi(g) \in A_t$; thus $g \in \mathcal{N}^{\text{ev}} \cap F_t[\check{Y}]$. By Lemma 3, $g = \sum_r c_r \nu_r(\check{Y})$ with $c_r \in A_t$. Multiplying back by the unit $\check{Y}^{-N}$,

$$f = \check{Y}^{-N} g = \sum_{r \geq 0} c_r \check{Y}^{-N} \nu_r(\check{Y}) \in \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} A_t \check{Y}^u \nu_r(\check{Y}) = S,$$

each summand being an A_t -multiple of the generator $\check{Y}^u \nu_r(\check{Y})$ with $u = -N$. Hence $\mathcal{N}^{\text{ev}} \subseteq S$, and with Lemma 2, $\mathcal{N}^{\text{ev}} = S$. $\square$

4 History

The lattice of Section 3 is the last term of a line that is additive before it is multiplicative and classical before it is quantum.

The classical case. Pólya's basis theorem: the polynomials taking integer values at every integer form a free $\mathbb{Z}$ -module on the binomial coefficients $\binom{x}{k}$, strictly larger than $\mathbb{Z}[x]$ [5, Prop. 1.1]. The subject it opened is surveyed in Cahen–Chabert [1].

The multiplicative case. Replacing the integers by a geometric progression $1, q, q^2, \dots$ gives the grid that appears here. For entire functions the analogue is Gel'fond's (1933), which Gramain records as the *analogue multiplicatif* of Pólya's theorem; for polynomials it is Gramain [3, Prop. 2.2], who exhibits a basis for the functions taking integral values at every q^n . One difference of setting matters: there q is a natural number ≥ 2 and integrality means values in $\mathbb{Z}$, whereas here q is an indeterminate and integrality means values in A_t . The grid is the same; the coefficient ring differs, and every structural consequence drawn here from the free parameter t is new with it.

The q -analogue. Harman–Hopkins [5, §1, Props. 1.1–1.2] deform the classical case: one variable, the additive grid $[n]_q$, coefficients in $\mathbb{Z}[q^{\pm 1}]$, no family parameter. The forward inclusion and the windowed reverse interpolation of Section 3 follow their §1. What is added here is the passage to the multiplicative grid $\check{Y} = q^x$ and to bilateral exponents $u \in \mathbb{Z}$, together with the Laurent clearing that derives the two-sided statement from the one-sided one. Their substitution $\check{Y} = 1 + (q - 1)x$ — which their §4 connects to the Cartan part of Lusztig's integral form — carries their grid node $x = [n]_q$ to $\check{Y} = q^n$ and their basis term to $\nu_k(\check{Y})$ term by term.

The quantum side. The ingredients are classical. The triangular decomposition and the freeness of the toral part go back to Lusztig [7, §3.2]; quantum Verma modules and their contravariant forms are treated in De Concini–Procesi [2, §§17.1–17.4]; over the fraction field the Harish–Chandra map is an isomorphism onto the invariants by De Concini–Procesi [2, §18.3] and Jantzen [6, Thm. 6.25 and §6.26], with Jantzen's §6.6 already placing the image inside the *even* toral invariants; and integral forms assembled, as here, from a divided-power negative part and a toral lattice appear in Habiro–Lê [4, §8G]. Clause (ii) rests on the corresponding GL_2 statement, proved earlier [9]. What none of them carries is the family parameter together with the integral condition: over the fraction field the centre is the invariant *monomial* ring, and the content of the theorem here is that over A_t it is the strictly larger Newton lattice.

Remark 1. *Clause (iii) is the quantum Pólya theorem of Harman and Hopkins, Quantum integer-valued polynomials (arXiv:1601.06110), in bilateral Laurent form. Their §1 Proposition (Prop. §1.2, **prop:qb inom**), the q -analogue of Pólya's 1919 theorem, states that the ring $\mathcal{R}_q^+ = \{P \in \mathbb{Q}(q)[x] : P([n]_q) \in \mathbb{Z}[q] \ \forall n \in \mathbb{N}\}$ is freely generated as a $\mathbb{Z}[q]$ -module by*

the q -binomial coefficient polynomials $\begin{bmatrix} x \\ k \end{bmatrix}_q$ (the q -deformed Newton basis on the additive grid $x = [n]_q$, with denominator $q^{\binom{k}{2}} [k]_q!$ — not the classical Pólya $\binom{x}{k}$); their §4 Proposition (Prop. §4.3, **prop:localization**), $\mathcal{R}_q = \mathcal{R}_q^+ \otimes_{\mathbb{Z}[q]} \mathbb{Z}[q^{\pm 1}]$, is the bilateral localization. In the current notation their parameter q coincides with our q ($= v^2$), so their displayed formulas read verbatim on our side.

The node-preserving affine substitution $\check{Y} = 1 + (q - 1)x$ is exactly their §4 z -variable; it carries the grid node $x = [n]_q$ to $\check{Y} = q^n$ and carries each basis term $\begin{bmatrix} x \\ k \end{bmatrix}_q$ onto $\nu_k(\check{Y})$ term by term — the factors $(q - 1)$ cancel in every Newton ratio. (They remark further that on adjoining square roots $K^2 = z$, $v^2 = q$ — which is precisely the passage to our T^+ and our v — the ring $\mathcal{R}_q[K, v]$ is equivalent to the Cartan part of Lusztig’s integral form of $U_v(\mathfrak{sl}_2)$. Nothing below rests on that remark.) The values-in- $\mathbb{Z}[q^{\pm 1}]$ side matches values-in- A_t with t a spectator, since evaluation is t -coefficient-wise and $\mathbb{Q}(q) \cap \mathbb{Z}[v^{\pm 1}] = \mathbb{Z}[q^{\pm 1}]$.

The two-variable (GL_2) form of this equivalence is established in [9]: there the grid criterion agrees with membership in the A_t -span of the product Newton generators, and passes to the full Laurent even lattice. The GL_1 statement above is the rank-one specialization of that transport.

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